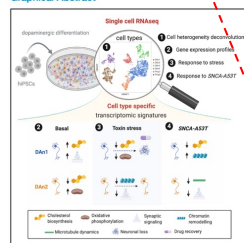


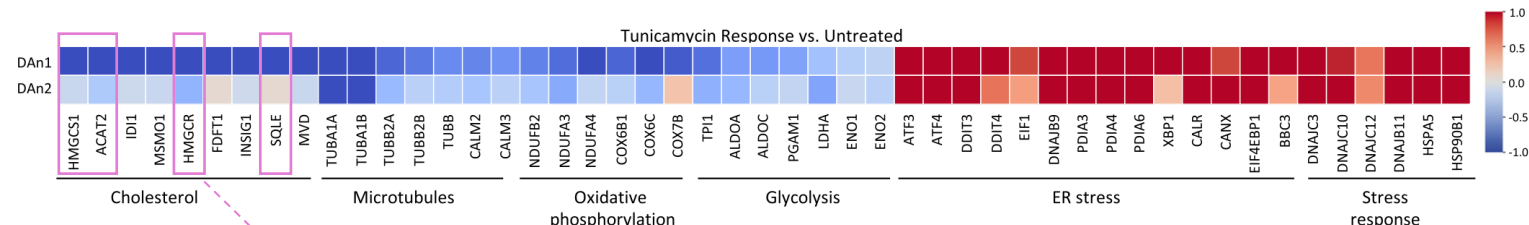
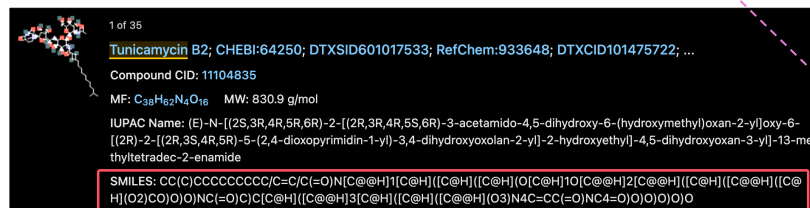
Graphical Abstract



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In Brief
Fernandes et al. identify and characterize cellular heterogeneity in human iPSC dopamine neurons using single-cell RNA-seq. The different subtypes of dopamine neurons illustrate distinct transcriptional profiles in response to toxic and genetic stressors. Furthermore, PD GWAS associated gene expression analysis is performed at the single-cell level.



After differentiation into dopaminergic neurons, cells were treated for 24h with either 5 μ M of Tunicamycin (Sigma) or 1 μ M of Rotenone (Sigma). For recovery studies, cells were pre-treated for 4 days with 100nM of Felodipine (Simga, replenished every two days), before stress treatments were initiated (Felodipine was kept during the duration of treatments).

pmid	title	perturbation_name	smiles	cell_type	dose	gene	regulation	source	original_statement
33053338	Single-Cell Transcriptomics of ...	tunicamycin	CCC...	DAn1 dopaminergic neuron	[5.0 μ M]	["HMGCS1", "ACAT2", "HMGCR", "SQLE"]	↓ Information on SQLE was available only in figures, necessitating multimodal data	Fig 4B, 4C	Top downregulated genes in response ...

Information on SQLE was available only in figures, necessitating multimodal data extraction and cross-validation.

To extend the impact of stress in human dopamine neurons, we treated dopamine cultures with tunicamycin, an inhibitor of N-linked glycosylation that results in ER stress. Analysis of top DEGs in response to tunicamycin in both DAN1 and DAN2 neurons revealed upregulation of multiple components of ER stress response (e.g., *CHOP/DDIT3*, *ATF4*, *PDIA4*, *CALR*), translation-

et al., 2003; Zhang et al., 2018). Top downregulated genes in response to tunicamycin and specific to DAN1 neurons were involved in cholesterol biosynthesis (e.g., *HMGCS1*, *ACAT2*, *HMGCR*) (Figures 4B and 4C). In addition, and common to DAN2, we found downregulation of genes involved with oxidative phosphorylation (e.g., *NDUFA4*, *COX7B*), microtubule dynamics

(A) Enrichment pathway analysis for top upregulated DEGs in DAN1 neurons in response to ER stress. Shown are also violin plots for *MANF* and *HSPA5*.

(B) Heatmap ($\log_2$ fold change gene expression) for main selected DEGs in response to ER stress for DAn1 and DAn2 neurons.

(C) Enrichment pathway analysis for top downregulated DEGs in DAN1 neurons in response to ER stress. Shown are also violin plots for *NDUFA4* and *SQLE*.

